

Chapter 7

Regularization for Deep Learning

A central problem in machine learning is how to make an algorithm that will perform well not just on the training data, but also on new inputs. Many strategies used in machine learning are explicitly designed to reduce the test error, possibly at the expense of increased training error. These strategies are known collectively as regularization. As we will see there are a great many forms of regularization available to the deep learning practitioner. In fact, developing more effective regularization strategies has been one of the major research efforts in the field.

Chapter 5 introduced the basic concepts of generalization, underfitting, overfitting, bias, variance and regularization. If you are not already familiar with these notions, please refer to that chapter before continuing with this one.

In this chapter, we describe regularization in more detail, focusing on regularization strategies for deep models or models that may be used as building blocks to form deep models.

Some sections of this chapter deal with standard concepts in machine learning. If you are already familiar with these concepts, feel free to skip the relevant sections. However, most of this chapter is concerned with the extension of these basic concepts to the particular case of neural networks.

In section 5.2.2, we defined regularization as “any modification we make to a learning algorithm that is intended to reduce its generalization error but not its training error.” There are many regularization strategies. Some put extra constraints on a machine learning model, such as adding restrictions on the parameter values. Some add extra terms in the objective function that can be thought of as corresponding to a soft constraint on the parameter values. If chosen carefully, these extra constraints and penalties can lead to improved performance

on the test set. Sometimes these constraints and penalties are designed to encode specific kinds of prior knowledge. Other times, these constraints and penalties are designed to express a generic preference for a simpler model class in order to promote generalization. Sometimes penalties and constraints are necessary to make an underdetermined problem determined. Other forms of regularization, known as ensemble methods, combine multiple hypotheses that explain the training data.

In the context of deep learning, most regularization strategies are based on regularizing estimators. Regularization of an estimator works by trading increased bias for reduced variance. An effective regularizer is one that makes a profitable trade, reducing variance significantly while not overly increasing the bias. When we discussed generalization and overfitting in chapter 5, we focused on three situations, where the model family being trained either (1) excluded the true data generating process—corresponding to underfitting and inducing bias, or (2) matched the true data generating process, or (3) included the generating process but also many other possible generating processes—the overfitting regime where variance rather than bias dominates the estimation error. The goal of regularization is to take a model from the third regime into the second regime.

In practice, an overly complex model family does not necessarily include the target function or the true data generating process, or even a close approximation of either. We almost never have access to the true data generating process so we can never know for sure if the model family being estimated includes the generating process or not. However, most applications of deep learning algorithms are to domains where the true data generating process is almost certainly outside the model family. Deep learning algorithms are typically applied to extremely complicated domains such as images, audio sequences and text, for which the true generation process essentially involves simulating the entire universe. To some extent, we are always trying to fit a square peg (the data generating process) into a round hole (our model family).

What this means is that controlling the complexity of the model is not a simple matter of finding the model of the right size, with the right number of parameters. Instead, we might find—and indeed in practical deep learning scenarios, we almost always do find—that the best fitting model (in the sense of minimizing generalization error) is a large model that has been regularized appropriately.

We now review several strategies for how to create such a large, deep, regularized model.

7.1 Parameter Norm Penalties

Regularization has been used for decades prior to the advent of deep learning. Linear models such as linear regression and logistic regression allow simple, straightforward, and effective regularization strategies.

Many regularization approaches are based on limiting the capacity of models, such as neural networks, linear regression, or logistic regression, by adding a parameter norm penalty $\Omega(\boldsymbol{\theta})$ to the objective function J . We denote the regularized objective function by $\tilde{J}$:

$$\tilde{J}(\boldsymbol{\theta}; \mathbf{X}, \mathbf{y}) = J(\boldsymbol{\theta}; \mathbf{X}, \mathbf{y}) + \alpha\Omega(\boldsymbol{\theta}) \quad (7.1)$$

where $\alpha \in [0, \infty)$ is a hyperparameter that weights the relative contribution of the norm penalty term, Ω , relative to the standard objective function J . Setting α to 0 results in no regularization. Larger values of α correspond to more regularization.

When our training algorithm minimizes the regularized objective function $\tilde{J}$ it will decrease both the original objective J on the training data and some measure of the size of the parameters $\boldsymbol{\theta}$ (or some subset of the parameters). Different choices for the parameter norm Ω can result in different solutions being preferred. In this section, we discuss the effects of the various norms when used as penalties on the model parameters.

Before delving into the regularization behavior of different norms, we note that for neural networks, we typically choose to use a parameter norm penalty Ω that penalizes *only the weights* of the affine transformation at each layer and leaves the biases unregularized. The biases typically require less data to fit accurately than the weights. Each weight specifies how two variables interact. Fitting the weight well requires observing both variables in a variety of conditions. Each bias controls only a single variable. This means that we do not induce too much variance by leaving the biases unregularized. Also, regularizing the bias parameters can introduce a significant amount of underfitting. We therefore use the vector $\mathbf{w}$ to indicate all of the weights that should be affected by a norm penalty, while the vector $\boldsymbol{\theta}$ denotes all of the parameters, including both $\mathbf{w}$ and the unregularized parameters.

In the context of neural networks, it is sometimes desirable to use a separate penalty with a different α coefficient for each layer of the network. Because it can be expensive to search for the correct value of multiple hyperparameters, it is still reasonable to use the same weight decay at all layers just to reduce the size of search space.

7.1.1 L^2 Parameter Regularization

We have already seen, in section 5.2.2, one of the simplest and most common kinds of parameter norm penalty: the L^2 parameter norm penalty commonly known as **weight decay**. This regularization strategy drives the weights closer to the origin¹ by adding a regularization term $\Omega(\boldsymbol{\theta}) = \frac{1}{2}\|\mathbf{w}\|_2^2$ to the objective function. In other academic communities, L^2 regularization is also known as **ridge regression** or **Tikhonov regularization**.

We can gain some insight into the behavior of weight decay regularization by studying the gradient of the regularized objective function. To simplify the presentation, we assume no bias parameter, so $\boldsymbol{\theta}$ is just $\mathbf{w}$. Such a model has the following total objective function:

$$\tilde{J}(\mathbf{w}; \mathbf{X}, \mathbf{y}) = \frac{\alpha}{2} \mathbf{w}^\top \mathbf{w} + J(\mathbf{w}; \mathbf{X}, \mathbf{y}), \quad (7.2)$$

with the corresponding parameter gradient

$$\nabla_{\mathbf{w}} \tilde{J}(\mathbf{w}; \mathbf{X}, \mathbf{y}) = \alpha \mathbf{w} + \nabla_{\mathbf{w}} J(\mathbf{w}; \mathbf{X}, \mathbf{y}). \quad (7.3)$$

To take a single gradient step to update the weights, we perform this update:

$$\mathbf{w} \leftarrow \mathbf{w} - \epsilon (\alpha \mathbf{w} + \nabla_{\mathbf{w}} J(\mathbf{w}; \mathbf{X}, \mathbf{y})). \quad (7.4)$$

Written another way, the update is:

$$\mathbf{w} \leftarrow (1 - \epsilon\alpha) \mathbf{w} - \epsilon \nabla_{\mathbf{w}} J(\mathbf{w}; \mathbf{X}, \mathbf{y}). \quad (7.5)$$

We can see that the addition of the weight decay term has modified the learning rule to multiplicatively shrink the weight vector by a constant factor on each step, just before performing the usual gradient update. This describes what happens in a single step. But what happens over the entire course of training?

We will further simplify the analysis by making a quadratic approximation to the objective function in the neighborhood of the value of the weights that obtains minimal unregularized training cost, $\mathbf{w}^* = \arg \min_{\mathbf{w}} J(\mathbf{w})$. If the objective function is truly quadratic, as in the case of fitting a linear regression model with

¹More generally, we could regularize the parameters to be near any specific point in space and, surprisingly, still get a regularization effect, but better results will be obtained for a value closer to the true one, with zero being a default value that makes sense when we do not know if the correct value should be positive or negative. Since it is far more common to regularize the model parameters towards zero, we will focus on this special case in our exposition.

mean squared error, then the approximation is perfect. The approximation $\hat{J}$ is given by

$$\hat{J}(\boldsymbol{\theta}) = J(\mathbf{w}^*) + \frac{1}{2}(\mathbf{w} - \mathbf{w}^*)^\top \mathbf{H}(\mathbf{w} - \mathbf{w}^*), \quad (7.6)$$

where $\mathbf{H}$ is the Hessian matrix of J with respect to $\mathbf{w}$ evaluated at $\mathbf{w}^*$. There is no first-order term in this quadratic approximation, because $\mathbf{w}^*$ is defined to be a minimum, where the gradient vanishes. Likewise, because $\mathbf{w}^*$ is the location of a minimum of J , we can conclude that $\mathbf{H}$ is positive semidefinite.

The minimum of $\hat{J}$ occurs where its gradient

$$\nabla_{\mathbf{w}} \hat{J}(\mathbf{w}) = \mathbf{H}(\mathbf{w} - \mathbf{w}^*) \quad (7.7)$$

is equal to $\mathbf{0}$.

To study the effect of weight decay, we modify equation 7.7 by adding the weight decay gradient. We can now solve for the minimum of the regularized version of $\hat{J}$. We use the variable $\tilde{\mathbf{w}}$ to represent the location of the minimum.

$$\alpha \tilde{\mathbf{w}} + \mathbf{H}(\tilde{\mathbf{w}} - \mathbf{w}^*) = 0 \quad (7.8)$$

$$(\mathbf{H} + \alpha \mathbf{I}) \tilde{\mathbf{w}} = \mathbf{H} \mathbf{w}^* \quad (7.9)$$

$$\tilde{\mathbf{w}} = (\mathbf{H} + \alpha \mathbf{I})^{-1} \mathbf{H} \mathbf{w}^*. \quad (7.10)$$

As α approaches 0, the regularized solution $\tilde{\mathbf{w}}$ approaches $\mathbf{w}^*$. But what happens as α grows? Because $\mathbf{H}$ is real and symmetric, we can decompose it into a diagonal matrix $\mathbf{\Lambda}$ and an orthonormal basis of eigenvectors, $\mathbf{Q}$, such that $\mathbf{H} = \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^\top$. Applying the decomposition to equation 7.10, we obtain:

$$\tilde{\mathbf{w}} = (\mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^\top + \alpha \mathbf{I})^{-1} \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^\top \mathbf{w}^* \quad (7.11)$$

$$= \left[\mathbf{Q} (\mathbf{\Lambda} + \alpha \mathbf{I}) \mathbf{Q}^\top \right]^{-1} \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^\top \mathbf{w}^* \quad (7.12)$$

$$= \mathbf{Q} (\mathbf{\Lambda} + \alpha \mathbf{I})^{-1} \mathbf{\Lambda} \mathbf{Q}^\top \mathbf{w}^*. \quad (7.13)$$

We see that the effect of weight decay is to rescale $\mathbf{w}^*$ along the axes defined by the eigenvectors of $\mathbf{H}$. Specifically, the component of $\mathbf{w}^*$ that is aligned with the i -th eigenvector of $\mathbf{H}$ is rescaled by a factor of $\frac{\lambda_i}{\lambda_i + \alpha}$. (You may wish to review how this kind of scaling works, first explained in figure 2.3).

Along the directions where the eigenvalues of $\mathbf{H}$ are relatively large, for example, where $\lambda_i \gg \alpha$, the effect of regularization is relatively small. However, components with $\lambda_i \ll \alpha$ will be shrunk to have nearly zero magnitude. This effect is illustrated in figure 7.1.

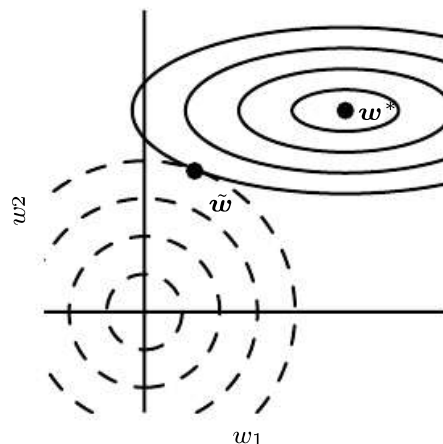


Figure 7.1: An illustration of the effect of L^2 (or weight decay) regularization on the value of the optimal $\mathbf{w}$. The solid ellipses represent contours of equal value of the unregularized objective. The dotted circles represent contours of equal value of the L^2 regularizer. At the point $\tilde{\mathbf{w}}$, these competing objectives reach an equilibrium. In the first dimension, the eigenvalue of the Hessian of J is small. The objective function does not increase much when moving horizontally away from $\mathbf{w}^*$. Because the objective function does not express a strong preference along this direction, the regularizer has a strong effect on this axis. The regularizer pulls w_1 close to zero. In the second dimension, the objective function is very sensitive to movements away from $\mathbf{w}^*$. The corresponding eigenvalue is large, indicating high curvature. As a result, weight decay affects the position of w_2 relatively little.

Only directions along which the parameters contribute significantly to reducing the objective function are preserved relatively intact. In directions that do not contribute to reducing the objective function, a small eigenvalue of the Hessian tells us that movement in this direction will not significantly increase the gradient. Components of the weight vector corresponding to such unimportant directions are decayed away through the use of the regularization throughout training.

So far we have discussed weight decay in terms of its effect on the optimization of an abstract, general, quadratic cost function. How do these effects relate to machine learning in particular? We can find out by studying linear regression, a model for which the true cost function is quadratic and therefore amenable to the same kind of analysis we have used so far. Applying the analysis again, we will be able to obtain a special case of the same results, but with the solution now phrased in terms of the training data. For linear regression, the cost function is

the sum of squared errors:

$$(\mathbf{X}\mathbf{w} - \mathbf{y})^\top (\mathbf{X}\mathbf{w} - \mathbf{y}). \quad (7.14)$$

When we add L^2 regularization, the objective function changes to

$$(\mathbf{X}\mathbf{w} - \mathbf{y})^\top (\mathbf{X}\mathbf{w} - \mathbf{y}) + \frac{1}{2}\alpha\mathbf{w}^\top \mathbf{w}. \quad (7.15)$$

This changes the normal equations for the solution from

$$\mathbf{w} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y} \quad (7.16)$$

to

$$\mathbf{w} = (\mathbf{X}^\top \mathbf{X} + \alpha \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}. \quad (7.17)$$

The matrix $\mathbf{X}^\top \mathbf{X}$ in equation 7.16 is proportional to the covariance matrix $\frac{1}{m} \mathbf{X}^\top \mathbf{X}$. Using L^2 regularization replaces this matrix with $(\mathbf{X}^\top \mathbf{X} + \alpha \mathbf{I})^{-1}$ in equation 7.17. The new matrix is the same as the original one, but with the addition of α to the diagonal. The diagonal entries of this matrix correspond to the variance of each input feature. We can see that L^2 regularization causes the learning algorithm to “perceive” the input $\mathbf{X}$ as having higher variance, which makes it shrink the weights on features whose covariance with the output target is low compared to this added variance.

7.1.2 L^1 Regularization

While L^2 weight decay is the most common form of weight decay, there are other ways to penalize the size of the model parameters. Another option is to use L^1 regularization.

Formally, L^1 regularization on the model parameter $\mathbf{w}$ is defined as:

$$\Omega(\boldsymbol{\theta}) = \|\mathbf{w}\|_1 = \sum_i |w_i|, \quad (7.18)$$

that is, as the sum of absolute values of the individual parameters.² We will now discuss the effect of L^1 regularization on the simple linear regression model, with no bias parameter, that we studied in our analysis of L^2 regularization. In particular, we are interested in delineating the differences between L^1 and L^2 forms

²As with L^2 regularization, we could regularize the parameters towards a value that is not zero, but instead towards some parameter value $\mathbf{w}^{(o)}$. In that case the L^1 regularization would introduce the term $\Omega(\boldsymbol{\theta}) = \|\mathbf{w} - \mathbf{w}^{(o)}\|_1 = \sum_i |w_i - w_i^{(o)}|$.

of regularization. As with L^2 weight decay, L^1 weight decay controls the strength of the regularization by scaling the penalty Ω using a positive hyperparameter α . Thus, the regularized objective function $\tilde{J}(\mathbf{w}; \mathbf{X}, \mathbf{y})$ is given by

$$\tilde{J}(\mathbf{w}; \mathbf{X}, \mathbf{y}) = \alpha \|\mathbf{w}\|_1 + J(\mathbf{w}; \mathbf{X}, \mathbf{y}), \quad (7.19)$$

with the corresponding gradient (actually, sub-gradient):

$$\nabla_{\mathbf{w}} \tilde{J}(\mathbf{w}; \mathbf{X}, \mathbf{y}) = \alpha \text{sign}(\mathbf{w}) + \nabla_{\mathbf{w}} J(\mathbf{X}, \mathbf{y}; \mathbf{w}) \quad (7.20)$$

where $\text{sign}(\mathbf{w})$ is simply the sign of $\mathbf{w}$ applied element-wise.

By inspecting equation 7.20, we can see immediately that the effect of L^1 regularization is quite different from that of L^2 regularization. Specifically, we can see that the regularization contribution to the gradient no longer scales linearly with each w_i ; instead it is a constant factor with a sign equal to $\text{sign}(w_i)$. One consequence of this form of the gradient is that we will not necessarily see clean algebraic solutions to quadratic approximations of $J(\mathbf{X}, \mathbf{y}; \mathbf{w})$ as we did for L^2 regularization.

Our simple linear model has a quadratic cost function that we can represent via its Taylor series. Alternately, we could imagine that this is a truncated Taylor series approximating the cost function of a more sophisticated model. The gradient in this setting is given by

$$\nabla_{\mathbf{w}} \hat{J}(\mathbf{w}) = \mathbf{H}(\mathbf{w} - \mathbf{w}^*), \quad (7.21)$$

where, again, $\mathbf{H}$ is the Hessian matrix of J with respect to $\mathbf{w}$ evaluated at $\mathbf{w}^*$.

Because the L^1 penalty does not admit clean algebraic expressions in the case of a fully general Hessian, we will also make the further simplifying assumption that the Hessian is diagonal, $\mathbf{H} = \text{diag}([H_{1,1}, \dots, H_{n,n}])$, where each $H_{i,i} > 0$. This assumption holds if the data for the linear regression problem has been preprocessed to remove all correlation between the input features, which may be accomplished using PCA.

Our quadratic approximation of the L^1 regularized objective function decomposes into a sum over the parameters:

$$\hat{J}(\mathbf{w}; \mathbf{X}, \mathbf{y}) = J(\mathbf{w}^*; \mathbf{X}, \mathbf{y}) + \sum_i \left[\frac{1}{2} H_{i,i} (\mathbf{w}_i - \mathbf{w}_i^*)^2 + \alpha |w_i| \right]. \quad (7.22)$$

The problem of minimizing this approximate cost function has an analytical solution (for each dimension i), with the following form:

$$w_i = \text{sign}(w_i^*) \max \left\{ |w_i^*| - \frac{\alpha}{H_{i,i}}, 0 \right\}. \quad (7.23)$$

Consider the situation where $w_i^* > 0$ for all i . There are two possible outcomes:

1. The case where $w_i^* \leq \frac{\alpha}{H_{i,i}}$. Here the optimal value of w_i under the regularized objective is simply $w_i = 0$. This occurs because the contribution of $J(\mathbf{w}; \mathbf{X}, \mathbf{y})$ to the regularized objective $\tilde{J}(\mathbf{w}; \mathbf{X}, \mathbf{y})$ is overwhelmed—in direction i —by the L^1 regularization which pushes the value of w_i to zero.
2. The case where $w_i^* > \frac{\alpha}{H_{i,i}}$. In this case, the regularization does not move the optimal value of w_i to zero but instead it just shifts it in that direction by a distance equal to $\frac{\alpha}{H_{i,i}}$.

A similar process happens when $w_i^* < 0$, but with the L^1 penalty making w_i less negative by $\frac{\alpha}{H_{i,i}}$, or 0.

In comparison to L^2 regularization, L^1 regularization results in a solution that is more **sparse**. Sparsity in this context refers to the fact that some parameters have an optimal value of zero. The sparsity of L^1 regularization is a qualitatively different behavior than arises with L^2 regularization. Equation 7.13 gave the solution $\tilde{w}$ for L^2 regularization. If we revisit that equation using the assumption of a diagonal and positive definite Hessian $\mathbf{H}$ that we introduced for our analysis of L^1 regularization, we find that $\tilde{w}_i = \frac{H_{i,i}}{H_{i,i} + \alpha} w_i^*$. If w_i^* was nonzero, then $\tilde{w}_i$ remains nonzero. This demonstrates that L^2 regularization does not cause the parameters to become sparse, while L^1 regularization may do so for large enough α .

The sparsity property induced by L^1 regularization has been used extensively as a **feature selection** mechanism. Feature selection simplifies a machine learning problem by choosing which subset of the available features should be used. In particular, the well known LASSO (Tibshirani, 1995) (least absolute shrinkage and selection operator) model integrates an L^1 penalty with a linear model and a least squares cost function. The L^1 penalty causes a subset of the weights to become zero, suggesting that the corresponding features may safely be discarded.

In section 5.6.1, we saw that many regularization strategies can be interpreted as MAP Bayesian inference, and that in particular, L^2 regularization is equivalent to MAP Bayesian inference with a Gaussian prior on the weights. For L^1 regularization, the penalty $\alpha\Omega(\mathbf{w}) = \alpha \sum_i |w_i|$ used to regularize a cost function is equivalent to the log-prior term that is maximized by MAP Bayesian inference when the prior is an isotropic Laplace distribution (equation 3.26) over $\mathbf{w} \in \mathbb{R}^n$:

$$\log p(\mathbf{w}) = \sum_i \log \text{Laplace}(w_i; 0, \frac{1}{\alpha}) = -\alpha \|\mathbf{w}\|_1 + n \log \alpha - n \log 2. \quad (7.24)$$